

Elektrodynamik Formelsamling

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Indhold

Indhold	1
1 Math	2
1.1 Vectors	2
1.2 Calculus	2
1.3 Fundamental theorem of calculus	3
2 Electrostatics	3
2.1 Electric field	3
2.2 Divergence and Curl	4
2.3 Electric potential	5
2.4 Work and Energy	6
2.5 Conductors	7
Capacitors	8
3 Potential	9
3.1 Laplace's equation	9

1 Math

1.1 Vectors

$$a(\bar{A} + \bar{B}) = a\bar{A} + a\bar{B}$$

$$\bar{A} \cdot \bar{B} = |\bar{A}||\bar{B}|\cos(\theta) \quad (1.1)$$

$$\bar{A} \times \bar{B} = |\bar{A}||\bar{B}|\sin(\theta)\hat{n} \quad (1.4)$$

$$(\bar{A} \times \bar{B}) = -(\bar{B} \times \bar{A}) \quad (1.6)$$

$$\bar{A} \cdot (\bar{B} \times \bar{C}) = \bar{B} \cdot (\bar{C} \times \bar{A}) = \bar{C} \cdot (\bar{A} \times \bar{B}) \quad (1.15)$$

$$\bar{A} \times (\bar{B} \times \bar{C}) = \bar{B}(\bar{A} \cdot \bar{C}) - \bar{C}(\bar{A} \cdot \bar{B}) \quad (1.17)$$

$$(\bar{A} \times \bar{B}) \cdot (\bar{C} \times \bar{D}) = (\bar{A} \cdot \bar{C})(\bar{B} \cdot \bar{D}) - (\bar{A} \cdot \bar{D})(\bar{B} \cdot \bar{C})$$

$$\bar{A} \times [\bar{B} \times (\bar{C} \times \bar{D})] = \bar{B}[\bar{A} \cdot (\bar{C} \times \bar{D})] - (\bar{A} \cdot \bar{B})(\bar{C} \cdot \bar{D}) \quad (1.18)$$

1.2 Calculus

$$dT = \left(\frac{\partial T}{\partial x} \hat{x} + \frac{\partial T}{\partial y} \hat{y} + \frac{\partial T}{\partial z} \hat{z} \right) \cdot (dx\hat{x} + dy\hat{y} + dz\hat{z})$$

$$= (\nabla T) \cdot d\vec{l}, \quad (1.35)$$

where the gradient (∇) is an operator:

$$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}. \quad (1.39)$$

The gradient is used to calculate the divergence of a vector with:

$$\nabla \cdot \bar{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}, \quad (1.40)$$

and the curl with:

$$\nabla \times \bar{v} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix}$$

$$= \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{x} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{y} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{z}. \quad (1.41)$$

The product rules for the gradient are:

$$\nabla(fg) = f\nabla g + g\nabla f$$

$$\nabla(\bar{A} \cdot \bar{B}) = \bar{A} \times (\nabla \times \bar{B}) + \bar{B} \times (\nabla \times \bar{A}) + (\bar{A} \cdot \nabla)\bar{B} + (\bar{B} \cdot \nabla)\bar{A}$$

$$\nabla \cdot (f\bar{A}) = f(\nabla \cdot \bar{A}) + \bar{A} \cdot (\nabla f)$$

$$\nabla \cdot (\bar{A} \times \bar{B}) = \bar{B} \cdot (\nabla \times \bar{A}) - \bar{A} \cdot (\nabla \times \bar{B})$$

$$\nabla \times (f\bar{A}) = f(\nabla \times \bar{A}) + \nabla f \times \bar{A}$$

$$\nabla \times (\bar{A} \times \bar{B}) = \bar{A}(\nabla \cdot \bar{B}) - \bar{B}(\nabla \cdot \bar{A}) + (\bar{B} \cdot \nabla)\bar{A} - (\bar{A} \cdot \nabla)\bar{B}$$

When we use the gradient to make a second derivative we get the Laplacian:

$$\nabla(\nabla T) = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}, \quad (1.42)$$

and the laplacian of a vector:

$$\nabla^2 \bar{v} = (\nabla^2 v_x) \hat{x} + (\nabla^2 v_y) \hat{y} + (\nabla^2 v_z) \hat{z}. \quad (1.43)$$

We also have the following identities:

$$\nabla \times (\nabla T) = 0 \quad (1.45)$$

$$\nabla \cdot (\nabla \times \bar{v}) = 0 \quad (1.46)$$

$$\nabla \times (\nabla \times \bar{v}) = \nabla(\nabla \cdot \bar{A}) - \nabla^2 \bar{A}. \quad (1.47)$$

1.3 Fundamental theorem of calculus

Gradient:

$$\int_a^{\bar{b}} (\nabla T) \cdot d\bar{l} = T(\bar{b}) - T(\bar{a}) \quad (1.55)$$

Divergence:

$$\int_V (\nabla \cdot \bar{v}) dV = \oint_S \bar{v} \cdot d\bar{a} \quad (1.56)$$

Curl:

$$\int_S (\nabla \times \bar{v}) \cdot d\bar{a} = \oint_P \bar{v} \cdot d\bar{l} \quad (1.57)$$

2 Electrostatics

2.1 Electric field

The force on a test charge (Q) from a single point charge (q), a distance $\bar{r}$ away, is calculated by Coulomb's law:

$$\bar{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{\bar{r}^2} \hat{r}, \quad (2.1)$$

where $\epsilon_0 = 8.85 \cdot 10^{-12} \frac{C^2}{N \cdot m^2}$ is the permittivity of free space.

If we have several point charges, at different distances from Q , we have an electric field:

$$\bar{F} = Q\bar{E}, \quad (2.3)$$

$$\bar{E} = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{\bar{r}_i^2} \hat{r}_i. \quad (2.4)$$

If we have a continuous charge distribution, we can calculate the electric field by:

$$\bar{E} = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r^2} \hat{r} dq, \quad (2.5)$$

where dq is the charge in a small element, be it line charge ($dq = \lambda dl$), area charge ($dq = \sigma da$) or volume charge ($dq = \rho d\tau$).

Special cases:

Field of infinite length wire:

$$E = \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{z}$$

Field of infinite plane:

$$\bar{E} = \frac{\sigma}{2\epsilon_0} \hat{a}$$

Field outside a uniformly charged sphere with total charge q , $r > R$:

$$\bar{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

2.2 Divergence and Curl

Field lines are imaginary lines used to represent the direction and strength of a field, such as an electric field. They start on positive charges and end on negative charges. The density of these lines indicates the field's strength; closer lines mean a stronger field.

Flux refers to the quantity of a field passing through a given surface. For an electric field, it is the product of the electric field strength and the area through which the field lines pass, considering the angle between the field lines and the surface.

$$\Phi = \int_S \bar{E} \cdot d\bar{a} \quad (2.11)$$

Gauss's law states that the total electric flux through a closed surface is proportional to the charge enclosed within that surface.

$$\oint \bar{E} \cdot d\bar{a} = \frac{Q_{\text{enc}}}{\epsilon_0}, \quad (2.13)$$

where Q_{enc} is the charge enclosed by the surface. It can be rewritten with the divergence theorem, when $Q_{\text{enc}} = \int_V \rho d\tau$:

$$\nabla \cdot \bar{E} = \frac{\rho}{\epsilon_0}. \quad (2.14)$$

To calculate the curl of the electric field we see that:

$$\int_{\bar{a}}^{\bar{b}} \bar{E} \cdot d\bar{l} = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_a} - \frac{q}{r_b} \right), \quad (2.18)$$

which on a closed interval gives us:

$$\oint \vec{E} \cdot d\vec{l} = 0, \quad (2.19)$$

$$\nabla \times \vec{E} = \vec{0}. \quad (2.20)$$

2.3 Electric potential

Since we have $\oint \vec{E} \cdot d\vec{l}$, it must be independent of the path taken, and therefore we can calculate the potential of a point as:

$$V(\vec{r}) = - \int_{\vec{O}}^{\vec{r}} \vec{E} \cdot d\vec{l}, \quad (2.21)$$

and the electric potential difference between two points as:

$$V(\vec{b}) - V(\vec{a}) = - \int_{\vec{a}}^{\vec{b}} \vec{E} \cdot d\vec{l}, \quad (2.22)$$

$$\Rightarrow \vec{E} = -\nabla V. \quad (2.23)$$

Poisson's equation:

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}, \quad (2.24)$$

which gives us a way to calculate the potential from the charge distribution:

For a collection of point charges:

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}, \quad (2.27)$$

For a continuous charge distribution:

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r} dq, \quad (2.28)$$

For a volume charge distribution:

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{r} d\tau'. \quad (2.29)$$

Special cases:

Potential of infinite length wire:

$$V = \frac{1}{4\pi\epsilon_0} \lambda \ln \left(\frac{b}{a} \right)$$

Potential of infinite plane:

$$V = \frac{\sigma}{2\epsilon_0} z$$

Potential of spherical shell:

$$V = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{q}{r} & r > R \\ \frac{1}{4\pi\epsilon_0} \frac{q}{R} & r < R \end{cases}$$

Figure (2.35) shows the way to go between electric field, electric potential and volume charge:

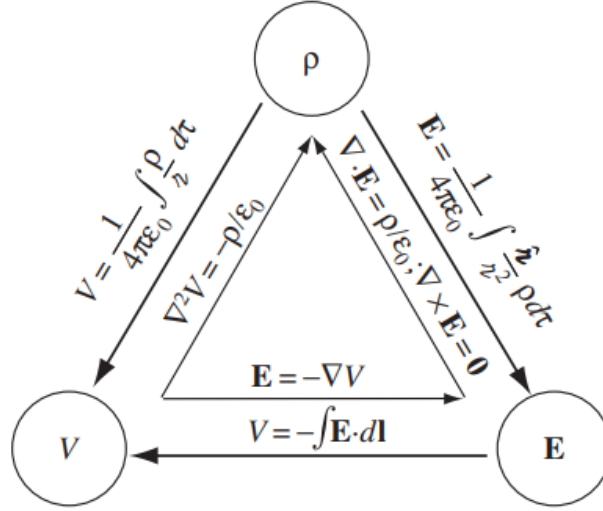


FIGURE 2.35

2.4 Work and Energy

If you want to move a test charge (Q) from point $\bar{a}$ to point $\bar{b}$, you have to do work against the electric field. The work done is the potential difference between the two points:

$$W = Q [V(\bar{b}) - V(\bar{a})]. \quad (2.38)$$

So if you want to move the point from infinity to a point $\bar{r}$, and you have the reference point in infinity:

$$W = QV(\bar{r}). \quad (2.39)$$

The work to move a collection of charges is:

$$W = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \sum_{j>i}^n \frac{q_i q_j}{r_{ij}}, \quad (2.40)$$

$$= \frac{1}{8\pi\epsilon_0} \sum_{i=1}^n \sum_{j \neq i}^n \frac{q_i q_j}{r_{ij}}, \quad (2.41)$$

$$= \frac{1}{2} \sum_{i=1}^n q_i V(\bar{r}_i). \quad (2.42)$$

For continuous charge distributions:

$$W = \frac{1}{2} \int V dq, \quad (2.43)$$

where again, dq is the charge in a small element, be it line charge ($dq = \lambda dl$), area charge ($dq = \sigma da$) or volume charge ($dq = \rho d\tau$).

If we take (2.43) for volume over all space, we get:

$$W = \frac{\epsilon_0}{2} \int E^2 d\tau. \quad (2.45)$$

2.5 Conductors

A conductor is a material where the electrons are free to move around. The electric field inside a conductor is zero, and the charges are distributed on the surface, like metal. The electric field inside a conductor is zero, and when it is placed in an electric field, the charges collect on the surface to create an electrical field inside the conductor, that cancels out the initial field.

$$\nabla \bar{E} = \frac{\rho}{\epsilon_0} = 0$$

Induced charges on a conductor are always on the surface, and the electric field is always perpendicular to the surface. For a conductor with a cavity, the electric field is zero inside the cavity, and the charges are on the surface of the cavity. Outside the conductor, the electric field is the same as if the conductor was not there.

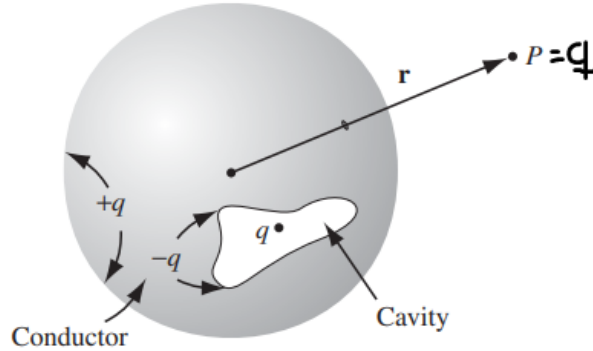


FIGURE 2.46

Because the field inside a conductor is zero, the field immediately outside is:

$$\bar{E} = \frac{\sigma}{\epsilon_0} \hat{n}, \quad (2.48)$$

The outward electrostatic pressure on the surface of a conductor is:

$$\bar{f} = \frac{1}{2\epsilon_0} \sigma^2 \hat{n}. \quad (2.51)$$

Capacitors

If we have two conductors with charge Q and $-Q$, the electric potential between them is:

$$V = V_+ - V_- = - \int_{(-)}^{(+)} \vec{E} \cdot d\vec{l},$$

if the shape is complicated, $\vec{E}$ is hard to calculate, but we know it is proportional to Q , and therefore V is also proportional, which gives us the capacitance:

$$C = \frac{Q}{V}. \quad (2.53)$$

We can talk about the capacitance of a single capacitor, this refers to (2.53) where the reference point for V is infinity.

For a parallel-plate capacitor the capacitance is:

$$C = \frac{A\epsilon_0}{d}, \quad (2.54)$$

To charge a capacitor you have to move the charge from the positive plate to the negative plate. the energy required to do this is:

$$W = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{1}{2} \frac{Q^2}{C}. \quad (2.55)$$

3 Potential

3.1 Laplace's equation

To easily calculate the potential in a region, we can use Laplace's equation:

$$\nabla^2 V = 0. \quad (3.4)$$

In one dimension the solution is:

$$V(x) = ax + b, \quad (3.6)$$

but for 2 dimensions it is not easy to calculate exactly, but you can calculate $V(x, y)$ as the average of the potential around the point:

$$V(x, y) = \frac{1}{2\pi R} \oint_{circle} V dl,$$

which might make no sense, but it can be used to calculate the charge distribution, when boundary conditions are known.

In three dimensions the value of V at point $\bar{r}$ is the average value of V over a spherical surface of radius R centered at $\bar{r}$:

$$V(\bar{r}) = \frac{1}{4\pi R^2} \oint_{sphere} V da.$$